

Supermart Grocery Sales Analysis Report

1. Overview

This project focuses on analyzing supermarket sales data to understand sales performance, customer demand, and business trends. Using data analysis and machine learning techniques, the project transforms raw data into meaningful insights.

Real-world connection: Retail businesses use similar analysis to improve inventory management, sales strategies, and profitability.

2. Objectives

- 1 Identify top-performing product categories
- 2 Analyze sales across regions and cities
- 3 Study monthly sales trends
- 4 Build predictive models for forecasting
- 5 Provide actionable business insights

3. Dataset

The dataset contains Order Date, Region, City, Category & Sub-Category, Product Name, Sales, Quantity, and Profit.

Dataset behavior: Meat & Fish category shows highest sales, sales increase over months, and regional variation exists.

Real-world comparison: Similar to supermarket billing system data.

4. Data Preprocessing

- 1 Converted Order Date into datetime format
- 2 Removed missing values and duplicates
- 3 Created new features: Day, Month, Year

Importance: Improves data quality, helps time-based analysis, and enhances model performance.

5. Exploratory Data Analysis (EDA)

Key findings: Meat & Fish is highest-selling category, sales increase over time, and some regions perform better.

Importance: Helps identify trends and supports business decisions.

6. Visual Analysis

- 1 Sales by Category – Meat & Fish dominates
- 2 Monthly Sales Trend – Increasing trend
- 3 Region-wise Sales – Some regions perform better
- 4 Sales vs Profit – Positive relationship
- 5 Correlation Heatmap – Strong correlation between sales and profit

7. Machine Learning Models

Linear Regression: Simple model, easy to understand but limited accuracy.

Random Forest: Advanced model with higher accuracy and better prediction performance.

8. Model Evaluation

Metrics used: Mean Squared Error (MSE) and R^2 Score.

Result: Random Forest performs better and provides more accurate predictions.

9. Business Impact (Real-World Case)

- 1 Problem: Overstocking, understocking, uneven sales, poor forecasting
- 2 Solution: Identify top products, predict demand, optimize inventory, improve pricing
- 3 Impact: Reduced waste, increased revenue, better forecasting, improved efficiency

10. Conclusion

This project demonstrates data preprocessing, EDA, visualization, machine learning, and business insights. It shows how data improves decision-making and business growth.